Question 18

Q: In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.

Theoretical Concept: Quantization in Medical Imaging

Medical images (like MRIs or CT scans) contain very subtle differences in tissue density. If the image is quantized to standard 8-bit (256 shades), a tumor and healthy tissue might map to the exact same digital gray value.

By using 12-bit or 16-bit quantization (4096 to 65536 shades), the system preserves the tiny radiometric differences crucial for a correct diagnosis.

💻 OpenCV Python Solution

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```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                              # Import Numpy for mathematical matrix operations
import matplotlib.pyplot as plt                                 # Import Matplotlib to visualize the output on screen

# Step 1: Simulate a 16-bit medical scan (range 0-65535)
medical_16bit = np.full((100, 100), 32000, dtype=np.uint16)  # Create a solid gray image array filled with a value
# Introduce a tiny tissue density anomaly (+10 intensity), invisible in 8-bit
cv2.circle(medical_16bit, (50, 50), 20, 32010, -1)  # Draw a filled or outlined circle on the image
```

```
# Step 2: Simulate destructive quantization by downcasting to 8-bit
medical_8bit = (medical_16bit / 256).astype(np.uint8)

# Step 3: Mathematically normalize the 16-bit slice to map the subtle data to a visible range
visual_enhancement = cv2.normalize(medical_16bit, None, 0, 255, cv2.NORM_MINMAX, dtype=cv2.CV_8U)  # Normalize pixe

# Step 4: Display the medical imaging necessity for high quantization
plt.figure(figsize=(8,4))  # Create a new figure window for display
plt.subplot(121), plt.imshow(medical_8bit, cmap='gray', vmin=0, vmax=255), plt.title('8-bit Quantized (Lost)')  # P
plt.subplot(122), plt.imshow(visual_enhancement, cmap='gray'), plt.title('16-bit Windowed (Found)')  # Place this i
plt.show()  # Show the final plot window to the user
```

🎯 Expected Output

The 8-bit image will be entirely solid gray, showing that the tumor data was destroyed by quantization. The 16-bit map clearly highlights the anomaly, proving the medical necessity of higher bit depths.

🚀 How to Execute & Submit

1. Google Colab Execution:

- Open Google Colab and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.